

**THE PROJECT FOR
OXYGEN MANIFOLD SYSTEM ESTABLISHMENT
IN 20 SELECTED UPAZILA HEALTH COMPLEXES
AND MEDICAL GAS CONNECTIVITY IN 4
SELECTED SCANU AT SELECTED HOSPITALS
IN
THE PEOPLE’S REPUBLIC OF BANGLADESH**

**PROJECT COMPLETION
REPORT**

APRIL 2023

**USAID’S MAMONI MNCSP: EMERGENCY
RESPONSES TO COVID 19 PROGRAM**

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LOCATION MAP OF 20 SLECTED UPAZILA HEALTH COMPLEXES

List of 20 Selected Upazila Health Complex

SL No	Facility Name	Upazila	District	Division
1	Shibchar Upazila Health Complex	Shibchar	Madaripur	Dhaka
2	Bhederganj Upazila Health Complex	Bhederganj	Shariatpur	
3	Naria Upazila Health Complex	Naria	Shariatpur	
4	Mathbaria Upazila Health Complex	Mathbaria	Pirojpur	Barishal
5	Nazirpur Upazila Health Complex	Nazirpur	Pirojpur	
6	Beani Baazar Upazila Health Complex	Beani Baazar	Sylhet	Sylhet
7	South Surma Upazila Health Complex	South Surma	Sylhet	
8	Zakiganj Upazila Health Complex	Zakiganj	Sylhet	
9	Kanaighat Upazila Health Complex	Kanaighat	Sylhet	
10	Kolmakanda Upazila Health Complex	Kolmakanda	Netrokona	Mymensingh
11	Barhatta Upazila Health Complex	Barhatta	Netrokona	
12	Tarakanda Upazila Health Complex	Tarakanda	Mymensingh	
13	Akhaura Upazila Health Complex	Akhaura	Brahmanbaria	Chattagram
14	Nabinagar Upazila Health Complex	Nabinagar	Brahmanbaria	
15	Hatiya Upazila Health Complex	Hatiya	Noakhali	
16	Begumganj Upazila Health Complex	Begumganj	Noakhali	
17	Sonaimuri Upazila Health Complex	Sonaimuri	Noakhali	
18	Faridganj Upazila Health Complex	Faridganj	Chandpur	
19	Kochua Upazila Health Complex	Kochua	Chandpur	Rajshahi
20	Shanthia Upazila Health Complex	Shanthia	Pabna	

TABLE OF CONTENTS

Table of Contents

Location Map of the Project

List of Appendics

Abbreviation

1. OUTLINE OF THE PROJECT.....	1
Background of the Project.....	1
Scope of Works as Critical Care Service	1
Project Period	2
Project Team and Stakholder	2
Project Planning Matrix	3
TA Plan for this Project	3
Project Activity Flow.....	5
Project Implementation Structure.....	6
2. SUMMARY OF MAUALS AND ACTION PLAN	8
Summary of Design Manuals of Oxygen Manifold System	8
Summary of Construction of Oxygen Manifold Room	8
Summary of Design Manual of Medical Gas Connectivity System for SCANU	8
3. PROJECT ACTIVITIES.....	10
Assessment of the Preliminary selected sites.....	10
Transfer of Knowledge and handover the sites.....	11
Layout of Manifold Room and Oxygen Pipeline Routes	12
Material Check and Inspection of Oxygen Manifold System.....	13
Civil Construction of Oxygen Manifold Room of 20 UHC	17
Electrical Works quality check along with earthening works	20
Pipeline brazing with Nitrogen purging	21
Pipeline System layout and Installation	23
Testing and Commissioning of Oxygen Manifold System.....	26
Commissioning of Oxygen Manifold System at 20 UHC	27
Commissioning of Medical Gas Connectivity of SCANUs	28
Operation and Maintenance Training at 20 UHC	34
4. CONCLUTIONS	37
Summery of Project Result.....	37
Findings and Recommendation of the Project	37
Application and Ackowledgement	39

LIST OF APPENDICES

Appendix-1:	Assessment of the site for 20 selected UHC	A-1
Appendix-2:	List of Material installed at the Project	A-2
Appendix-3:	Testing and Commissioning Reports of 20 UHC	A-3
Appendix-4:	Installation Records of the Project.....	A-4
Appendix-5:	As build Drawings of the projects.....	A-5
Appendix-6:	List of Training at facilities	A-6
Appendix-7:	Tour reports of the Project	A-7

ABBREVIATIONS

BNBC	Bangladesh National Building Code
BNPRP	National Preparedness and Response Plan
BOQ	Bill of Quantity
CRM	Climate Risk Management
COVID 19	Coronavirus Disease
DGHS	Director General of Health Service
EMMP	Environmental Monitoring and Mitigation Plan
GOB	Government of Bangladesh
GEM	Ground Enhancement Material
HED	Health Engineering Department
HTM-2022	Health Technical Memorendum
JICA	Japan International Cooperation Agency
LGED	Local Government Engineering Department
Mamoni MNCSP	Mamoni Maternal and Newborn Care Strenthening Project
MOH&FW	Mintry of Health and Family Welfare
MP	Member of Pairliament
NFPA-99	National Fire Protection Association
NGO	Non Government Organization
NEMEMW	National Electro-Medical Equipment Maintenance Workshop
O&M	Operationa & Maintenance
PPE	Personal Protective Equipment
PWD	Public Works Department
RMO	Resident Medical Officer
SCI	Save the Children International
SCANU	Special Care Newborn Unit
SIL	Spectra International Limited
TOR	Terms of Referance
TWG	Technical Working Group
USAID	United States Agency of International Development
UHC	Upazila Health Complex
UH&FPO	Upazila Health & Family Planning Officer
UNICEF	United Nations Children's Fund

1. OUTLINE OF THE PROJECT

Background of the Project:

The Bangladesh government has developed a National Preparedness and Response Plan (BNPRP) to prevent and control the COVID-19 pandemic situation, and international NGOs are supporting the government in implementing the priority actions set in the plan. USAID's MaMoni MNCSP: Emergency responses to COVID 19 pandemic project aims to strengthen the health system's response to community transmission of COVID-19 and minimize health risks to individuals. Save the Children International is managing the project's response plan, which supports the MOH&FW to strengthen the health system's capacity to treat and manage COVID-19 cases, improve infection prevention and control practices, support community-based interventions to improve case detection and management, and enhance the functions of the national call center in documentation and reporting.

The Consultant, Critical Care Service, will facilitate the project's activities related to monitoring and supervision of the installation, commissioning, and maintenance of the Central Medical Gas Pipeline System by Manifold in 20 selected Upazila Health Complexes of Bangladesh to strengthen the oxygen ecosystem as part of the health system support. The consultant will coordinate with the organization installing the Manifold system, engage with the government and relevant stakeholders to provide technical assistance, and work in collaboration with the hired firm/individual technical expert and program team. The consultant will also be engaged to undertake prompt actions per the recommendation of USAID and the Government of Bangladesh on emerging issues and changing COVID-19 pandemic situations.

Scope of Works as Consultant-Critical Care Service:

USAID's MaMoni MNCSP: COVID-19 Pandemic response team under the guidance of the Project Director, the Consultant- Critical Care Service scope of work was as follows,

1. As the lead consultant, I will oversee the installation of the Central Medical Gas Pipeline System by manifold, ensuring compliance with HTM-2022/NFPA-99 and local codes, standards, rules, regulations, and laws. This includes supporting physical structural analysis for selected UHC, assisting in designing and finalizing technical specifications, and monitoring site supervision and quality assurance based on approved design and technical specifications.
2. I will also oversee renovation work for the manifold room based on site assessment reports and preventive maintenance, trouble-shooting, and repair-maintenance. Additionally, I will monitor the training of hospital staff, including on-site training, and identify challenges of implementation with tentative mitigation plans, promptly notifying Save the Children.
3. I will be responsible for developing and operationalizing site supervision and quality assurance in coordination with the hired firm/individual technical expert, and coordinating with relevant stakeholders such as MOHFW, DGHS, HED, PWD, NEMEMW, and Save the Children to ensure smooth implementation of the project. This includes coordinating any renovation work needed, facilitating training plans for hospital staff, supervising additional oxygen refilling and supply to proposed 20 facilities, and accompanying the hired firm/individual technical expert for quality assurance of monthly maintenance plans and

inspections.

4. I will cross-check the provided weekly progress report and attend monthly progress presentations to be shared in progress review meetings. Additionally, I will supervise junior consultants and other consultants related to the project as the lead consultant.

Project Period:

Timeline to complete this project was 12 July 2022 to 31 March 2023. As a Consultant I was full time committed technical team leader to complete the project work within this timeframe, with frequent field movement.

The project time as extended as per work condition of the vendor's work. Such as, we have waiting until all digital Alarms were installed in all sites. Alarms were not installed 11 sites among 20 sites due to not availability at vendor's stock.

Moreover, the as built drawing where not prepared according to the standard also need to extend the project work almost a month.

Project Team and Stakeholder:

Teams and Stakeholders of the Medical Oxygen Manifold System Project (Implemented by Save the Children International in 20 Upazila Health Complex in Bangladesh and funded by USAID)

1. Save the Children International Project Team: This team is responsible for the overall management and implementation of the project. It includes project managers, engineers, technicians, and other relevant personnel who are responsible for tasks such as needs assessment, procurement, contractor selection, system design, installation, training, monitoring, documentation, and reporting.
2. Upazila Health Complex Staff: The staff of the 20 Upazila Health Complexes is key stakeholders in the project. They play a crucial role in the successful implementation of the manifold system as they are responsible for operating, maintaining, and troubleshooting the system on a day-to-day basis. This includes roles such as doctors, nurses, biomedical engineers, maintenance personnel, and other relevant staff.
3. Patients and Communities: The patients, who receive medical treatment at the 20 Upazila Health Complexes, as well as the communities in the respective upazilas, are stakeholders in the project. The reliable supply of medical oxygen from the manifold system directly impacts patient care and outcomes, and the benefits of the system will be received by the general people of the respective upazilas.
4. Management and Administration of Upazila Health Complexes: The management and administration of each Upazila Health Complex are stakeholders in the project, as they are responsible for providing the necessary support, resources, and approvals for the successful implementation of the manifold system. This includes hospital administrators, facility managers, procurement officers, and other relevant personnel.
5. Contractors and Suppliers: The contractors and suppliers responsible for the supply, installation, and commissioning of the manifold system are stakeholders in the project. They play a crucial role in providing the necessary equipment, ensuring its quality, compatibility, and timely delivery, and providing technical support during installation and commissioning.
6. USAID: As the funding agency for the project, USAID is a key stakeholder. They provide the financial resources for the project and may have specific requirements and expectations related to the project's implementation, reporting, and outcomes.

7. **Regulatory Authorities:** Regulatory authorities, such as local health departments and relevant government agencies in Bangladesh, are stakeholders in the project. They may have specific regulations, guidelines, and approvals that need to be complied with during the implementation of the manifold system.
8. **Save the Children International Country Office:** The Save the Children International Country Office in Bangladesh is a stakeholder in the project, as they provide overall guidance, support, and oversight for the project's implementation.
9. **Other Stakeholders:** Other stakeholders may include local community members, patient advocacy groups, NGOs, and other relevant organizations or individuals who have an interest in the project's outcomes and impact on healthcare services in the community.

Effective coordination, communication, and engagement with all these teams and stakeholders are essential for the successful implementation of the Medical Oxygen Manifold System Project. Regular communication, stakeholder engagement, and addressing concerns or issues raised by stakeholders are key aspects of the project management approach.

Project Planning Matrix:

Goals and Objectives	Scope	Timeline	Roles and Responsibilities	Budget and Resources	Risks and Mitigation Strategies	Monitoring and Evaluation
Provide reliable and sustainable Oxygen Manifold System to ensure connectivity of oxygen to 20 selected UHC and 4 SCANU at different Hospitals in Bangladesh	Design, procurement, installation, and testing of oxygen connectivity systems; training of hospital staff	12 Month	USAID: funding and technical oversight; Save the Children International: project implementation and coordination; Contractor: system design, procurement, installation, testing and commissioning	USAID: \$X; Save the Children International: 1: in-kind resources; Contractor: \$Y	Risks: (1) Delays in procurement or installation, (2) Technical challenges, (3) Global Economic instability; (4) Changes of Import policy of Bangladesh. Mitigation Strategies: (1) Establish clear communication channels, (2) Identify and address technical issues early, (3) Work with local stakeholders to build required support	Data collection on system performance and hospital outcomes; regular reporting to USAID and Save the Children International

TA Plan for this project:

As a consultant for the Medical Gas Connectivity for SCANU project, a proper Technical Assistance (TA) plan could include the following:

1. **Initial Assessment:** Conducting a comprehensive assessment of the existing oxygen manifold system in the 20 UHCs to identify any gaps, challenges, or areas for improvement. This assessment would involve reviewing the system's design, installation, operation, maintenance, and compliance with standards and guidelines.
2. **Gap Analysis:** Analyzing the findings from the initial assessment to identify the gaps and challenges in the current oxygen manifold system, including any technical, operational, or training-related deficiencies. This analysis would help in identifying the areas that require technical assistance and support.

3. Customized Training: Developing and implementing a customized training plan for the UHCs' healthcare personnel, including junior mechanics, ward boys, senior staff nurses, and doctors, specifically focusing on the operation and maintenance of the oxygen manifold system, digital alarms, line ball valves, outlets, flow meters, and humidifiers. The training would also cover emergency response procedures, compliance with standards, and documentation requirements.
4. Standard Operating Procedures (SOPs): SOPs are as per standard practice to operate the Automatic and Manual manifold. And this unit's operation manual should be provided by the manufacturer themselves. And it should be provide by the vendors uring the training and after installation and commissioning of the systems. Developing and implementing standardized operating procedures for the operation, maintenance, and monitoring of the oxygen manifold system in the UHCs. The SOPs of the medical gas pipeline can be followed the recent BNBC that published on 2020, provide step-by-step guidelines for healthcare personnel to follow in their daily operations and ensure consistency in the system's operation across all UHCs.
5. Compliance Monitoring: Establishing a system for regular monitoring and auditing of the oxygen manifold system's compliance with relevant standards such as HTM 2022 or NFPA 99, guidelines from BNBC, and local and international regulations. This would involve conducting regular audits, inspections, and checks to ensure that the system is being operated and maintained in compliance with the established SOPs and regulatory requirements.
6. Emergency Preparedness: Developing and implementing emergency preparedness plans for unforeseen interruptions in the oxygen supply, including backup options, contingency plans, and emergency response protocols. This would ensure that healthcare personnel are well-prepared to handle any emergencies related to the oxygen manifold system and can respond promptly and effectively.
7. Documentation and Reporting: Establishing a system for proper documentation and reporting of the oxygen manifold system's operation, maintenance, and monitoring activities. This would include maintaining records of pressure readings, flow meter readings, alarm logs, maintenance schedules, and any deviations or issues encountered. Regular reporting to project stakeholders and management would be essential to ensure transparency and accountability.
8. Continuous Improvement: Implementing a system for continuous improvement of the oxygen manifold system's performance based on feedback from healthcare personnel, stakeholders, and ongoing monitoring. This would involve identifying areas for improvement, implementing corrective actions, and evaluating the effectiveness of the implemented measures.
9. Regular Follow-up and Support: Providing regular follow-up and ongoing support to the UHCs, including site visits, troubleshooting assistance, and technical guidance, to ensure the sustained operation and maintenance of the oxygen manifold system. This would also involve addressing any new challenges or issues that may arise during the project's implementation.
10. Knowledge Transfer: Ensuring effective knowledge transfer to the UHCs' healthcare personnel by providing comprehensive training materials, user manuals, and documentation for reference. This would enable the UHCs to develop in-house capabilities

for the proper operation and maintenance of the oxygen manifold system beyond the project's duration.

Overall, the TA plan would aim to ensure the efficient and reliable operation of the oxygen manifold system in the 20 UHCs, enhance healthcare personnel's knowledge and skills, and promote compliance with standards and guidelines, ultimately improving patient care and outcomes in newborn care units

Project Activity Flow:

The Project activity flow for the Oxygen connectivity project in Bangladesh, with the Upazila Health Complexes and the Director General of Health Service of Bangladesh government as the end users, facing the required oxygen connectivity during the COVID-19 pandemic:

1. Project Initiation Phase (Month 1)
 - The Director General of Health Service of Bangladesh government identifies the need for oxygen connectivity in Upazila Health Complexes to address the COVID-19 pandemic.
 - Save the Children International (SCI) is selected as the implementer and hires the technical consultants in separate different phases to assist with system design and procurement, monitoring and supervising the installation and commissioning of the system
 - The project team develops the project planning matrix, which outlines the goals, scope, timeline, roles and responsibilities, budget, risks and mitigation strategies, and monitoring and evaluation plan for the project.
2. Design and Procurement Phase (Months 2-4)
 - The technical consultant works with the project team to design the oxygen connectivity systems, including determining the appropriate equipment, specifications, and installation requirements.
 - The project team prepares a detailed procurement plan, which includes identifying potential suppliers, obtaining quotes, and selecting the most appropriate vendors based on technical and financial criteria.
 - SCI works with the relevant authorities to obtain the necessary permits and approvals for the procurement and installation of the oxygen connectivity systems.
3. Assessment, Installation and Testing Phase (Months 5-9)
 - The technical consultant along with contractor TWG joint visit the site to select the Oxygen Manifold room site and the oxygen outlets distribution as per requirement of each UHC.
 - The contractor procures the necessary equipment and materials, with oversight and approval from SCI and the relevant authorities.
 - The contractor begins the manifold room construction in the first 30 days to 45 days
 - Then the contractor starts the installation of the oxygen connectivity systems in the Upazila Health Complexes, following the design and installation specifications provided by the technical consultant, which is followed as per the standard of HTM

2022 and NFPA 99 guideline.

- The technical consultant provides oversight and technical support to the contractor during the installation and testing phase, ensuring that the systems are installed properly and meet the required specifications and as per the standard of HTM 2022 and NFPA 99.
- The project team develops a training plan for hospital staff, which includes training on how to use and maintain the oxygen connectivity systems.
- 4. Training and Capacity Building Phase (Months 10)
 - The project team conducts training sessions for hospital staff, with support from the technical consultant and the contractor.
 - The project team works with hospital staff to develop standard operating procedures for the use and maintenance of the oxygen connectivity systems.
 - The project team conducts a final review and testing of the systems to ensure that they are functioning properly and meeting the required specifications.
- 5. Project Closeout Phase (Month 11-12)
 - The project team prepares a final report on the project, which includes an evaluation of the project outcomes and lessons learned.
 - The project team coordinates with the relevant authorities to ensure that the oxygen connectivity systems continue to be used and maintained properly.
 - The project team provides handover ceremonies to transfer the ownership and responsibility of the oxygen connectivity systems to the Upazila Health Complexes and the relevant authorities.

Overall, the activity flow places greater emphasis on obtaining the necessary permits and approvals from the relevant authorities, and working closely with them throughout the project implementation phase to ensure that the oxygen connectivity systems meet the specific needs and requirements of the Upazila Health Complexes and SCANU of 4 Hospitals of DGHS of Bangladesh Governments to fight against the global pandemic like COVID-19.

Project Implementation Structure:

The Project Implementation Structure for the Oxygen Manifold System establishment Project for 20 UHC and Medical Gas Connectivity of 4 SCANU at different hospitals in Bangladesh would be as follows:

1. Project Director:
 - The Project Director was responsible for overall project oversight and decision-making.
 - The Project Director was employed by Save the Children International (SCI) and will be the primary point of contact with the donor, USAID.
2. Project Coordinator:
 - The Project Coordinator was responsible for managing the day-to-day implementation of the project and ensuring that it is delivered on time, within budget, and to the required quality standards.
 - The Project Coordinator was employed by SCI and will report to the Project Director.

3. Technical Consultants:

- Primarily one consultant has been selected to assist the procurement work. He prepares the technical documentation, such as Tender Documents with BOQ (Bill of Quantity) and detail technical specification to invite the tender. The selection criteria and the preliminary assessment of the sites also prepared by him.
- Later on two Technical Consultants, Consultant – Critical Care Service and Consultant – Oxygen Eco System were selected for field work monitoring and supervising on behalf of Save the Children International. They are responsible for providing technical expertise and guidance to the project team.
- The Technical Consultants were employed by SCI and will work closely with the Project Coordinator and the contractor.

4. Contractor Team:

- The Contractor Team was responsible for procuring and installing the necessary equipment and materials for the oxygen connectivity systems in the Upazila Health Complexes and 4 SCANU at 4 Government Hospitals.
- The Contractor Team comprised the General Manager, Assistant General Manager, and Senior Managers of the contractor company selected by SCI through a competitive bidding process.

5. Project Team:

- The Project Team was responsible for supporting the Project Coordinator in the day-to-day management of the project implementation.
- The Project Team comprised staff from SCI and the contractor.

6. Hospital Staff:

- Hospital Staff was responsible for operating and maintaining the oxygen connectivity systems once they are installed.
- The Project Team provides training and capacity building to hospital staff to ensure that they are able to use and maintain the systems properly.

Overall, this implementation structure ensures that there was a clear leadership and accountability for the project, and that there were a team of experts with the necessary technical and managerial skills to deliver the project successfully.

2. SUMMARY OF MAUALS AND ACTION PLAN

Summary of Design Manual of Oxygen Manifold System:

The Design Manual of Medical Gas Pipeline System outlines the requirements for the installation of a medical gas pipeline system in 20 selected Upazila Health Complexes (UHC) in Bangladesh. The system will provide oxygen as the only medical gas, with a total of 40 number outlets across the 20 UHCs. The system will use an oxygen manifold system with 36 number of 5.8 m³ oxygen cylinder. The system will be designed and installed in accordance with the Healthcare Technical Memorandum (HTM) 2022 standard.

The manual covers the technical specifications for the system components, including the pipeline network, manifold systems (both Automatic and Manuals), cylinder bank, alarms, valves and other safety features.

It also outlines the installation requirements and procedures, testing and commissioning protocols, and ongoing maintenance and safety management procedures. The manual emphasizes the importance of safety and reliability in the design and installation of the medical gas pipeline system, to ensure that patients receive the medical gas they need in a timely and safe manner.

Summary of Construction Guideline of Oxygen Manifold Room:

General guidelines for the design of Oxygen Manifold Rooms include:

- The room should be well-ventilated to prevent the buildup of oxygen and minimize the risk of fire.
- The room should be free of flammable materials such as fabrics, papers, or oils.
- The room should be equipped with an oxygen monitoring system that continuously monitors the oxygen concentration and alerts staff if the concentration becomes too high.
- The room should have a backup oxygen supply in case of a primary supply failure.
- The room should be designed to meet all applicable codes (BNBC) and standards for healthcare facilities (HTM 2022/NFPA 99).
- As this room is not in the main building and separately provide the Lightning arrestor along with Earthening system.

Summary of Design Manual of Medical Gas Connectivity System for SCANU:

The Design Manual of Medical Gas Connecting System for SCANU for the installation of a medical gas pipeline system at different hospital in Bangladesh is as per MTM 2022 (Health Technical Memorendum).

The requirement of the 4 SCANU is differ from each of the facility.

In Osmani Medical College Hospital SCANU, there are 39 sets of Oxygen outlets (NFPA 99 standard Ohmeda style) already installed but no Medical Air and Vaccum outlets. In Sunamgonj 250 Bed Sadar Hospital they have 14 numbers of Oxygen and Vaccum Outelets

but no Medical Air outlets. Where as in Habiganj and Manikganj 250 Sadar Hospital SCANU's there were no medical gas outlets. Two newly SCANU has been building up without any outlets of medical gases. But all the hospital has existing medical gas pipeline systems. So, all of those SCANU's required the medical gas connectivity with existing systems with line ball valves for each gases and premises. Beside the connectivity of SCANU of Habiganj Sadar Hospital Oxygen and Vacuum outlets has also install in Maternity ward, Labour Room, and Labour Observation Wards to provide maximum utilization of the facility.

Total 100 outlets installed in each SCANUs of Osmani Medical College Hospital, Habiganj 250 Bed Sadar Hospital and Manikganj 250 Bed Sadar Hospital. And only 14 numbers of Medical Air outlets at Sunamganj Sadar Hospital SCANU.

But the main draw back of the system is there are no monitoring systems or alarms recommended by the authority or previous consultant. As a result the availability of the Oxygen or other gases could not identify and may cause fatal injury of the new born babies.

To ensure smooth installation of the outlets we recommend Bed Head Unit at all SCANUs.

3. PROJECT ACTIVITIES

Need Assessment of the Preliminary selected sites:

After conducting visits to 29 Upazila Health Complexes (UHCs), the selection of 20 UHCs for the installation of the Oxygen Manifold system was made. The assessment process involved meeting with the Civil Surgeon initially, followed by visits to the UH&FPO and/or RMO of each facility. The selection criteria were as follows:

- Recommended listed UHC from DGHS, where an oxygen system was not already installed.
- The need for strengthening oxygen therapy for child and maternity care, requiring installation of an Oxygen Manifold system.
- Severe impact of the COVID-19 pandemic in 2020-2021 due to inadequate oxygen supply.
- Remote Upazilas, such as islands, where instant oxygen supply is not feasible.
- Availability of adequate manpower to maintain the oxygen manifold system at the UHC.
- Site assessment of the UHC, including the selection of a Manifold Room with loading and unloading facilities for oxygen cylinders.
- The UHC should have a minimum of 100 bed facilities, as 40 oxygen outlets were to be installed. The positioning of these outlets was decided through discussions with doctors and senior staff nurses.

Based on these criteria, the 20 UHCs were selected after consultation with the Civil Surgeon, UH&FPO, and/or RMO, and a joint visit with the vendor TWG was conducted prior to the start of the project work.

Based on the factors and status finding after initial assessment, it appears that there are some challenges and considerations that need to be addressed during the project implementation. These include:

- I. Building structure demolition and reconstruction: Some UHCs have structures that are slated for demolition and reconstruction in the near future. In such cases, it is wise to follow the recommendations of the Civil Surgeon or UH&FPO and not install the Oxygen Manifold system on those structures, as it may not be practical or cost-effective.
- II. Existing oxygen lines installed without proper supervision: Some UHCs have existing oxygen lines installed during the COVID-19 pandemic, but without proper supervision from relevant authorities such as JICA, LGED, USAID, or UNICEF. It is important to assess the quality and safety of these existing lines before proceeding with the installation of the Oxygen Manifold system to ensure that they meet the required standards.
- III. Lack of safety protocols and maintenance: Some UHCs have facilities where oxygen is being used but without adherence to safety protocols such as HTM 2022, BNBC, or NFPA 99. It is crucial to ensure that proper safety protocols are followed during the installation and operation of the Oxygen Manifold system to ensure the safety of patients, staff, and the facility. Additionally, maintenance protocols should be established to ensure the longevity and optimal functioning of the system.
- IV. Location of UHCs and recommendations of local MP: Some UHCs are located in areas prone to flooding or have recommendations from local MPs for installation of the Oxygen Manifold system despite concerns raised by the Civil Surgeon or UH&FPO. It is important to carefully assess the feasibility and suitability of the location based on factors such as flood risk, infrastructure stability, and long-term sustainability of the project.

Considering these factors and the status of the UHCs, it is essential to work closely with the Civil Surgeon, UH&FPO, and other relevant stakeholders to address any challenges, ensure adherence to safety protocols, and ensure the successful installation and operation of the Oxygen Manifold system in these UHCs. Regular monitoring and evaluation should also be conducted to ensure the ongoing effectiveness and safety of the system.

Transfer of Knowledge and handover the sites:

Spectra International Limited (SIL) was awarded the contract for the installation of the Oxygen Manifold system in the 20 selected UHCs, along with the 4 SCANU medical gas connectivity work. It's important to communicate and share the assessment report and site location layout drawings with Save the Children, as it helps in ensuring transparency and coordination among all stakeholders involved in the project.

Sharing the assessment report with Save the Children allows them to have a comprehensive understanding of the status of the UHCs and the challenges identified during the assessment process. This information can help them align their support and interventions accordingly to address any potential issues during the implementation phase.

Providing site location layout drawings for each of the 20 UHCs' oxygen manifold rooms is also crucial, as it helps Save the Children visualize the planned setup and understand the requirements for installation. This can facilitate their coordination with SIL and other stakeholders involved in the project to ensure smooth implementation and adherence to the planned design.

Effective communication and coordination among all parties involved, including SIL, Save the Children, UHCs, and other stakeholders, are vital for the successful installation of the Oxygen Manifold system in the selected UHCs. Regular updates, discussions, and coordination meetings can help address any challenges, clarify requirements, and ensure that the project progresses as planned.

We plan to provide them training along with their managers to maintain the work quality standard and safety protocol. We had visit the vander's office and meet with project implementation team. A presentation on medical gas pipeline work with safety instruction and EMMP and CRM protocol of Save the Children has ben informed. And ask them to maintain the protocol strictly. But as all the civil construction is very small volume and site were mostly at remote Upazila. The vander select the subcontractor for civil construction locally, hence they were not used to maintain the standard level of PPE protocol during their construction works.

The work plan from SCI consultant was given to SIL and asks them to start the work of construction of manifold room and copper pipline bazing works simulteneously. But the SIL recommeded to ensure the team work with optimum management of the project work with fully secured the delivered materials, they need to build up the manifold room first and provide the security of those material they can ensure the copper pipe brazing work as per scheduled time. SCI accept the logical explanation of the vander and revised the plan as per their required time frame.

A joint visit to the selected 20 UHC has scheduled on that meeting to ensure the hand over process. A two member of our consultant team devided in two separate groups with the vandors project managers to visit the sites.

We introduce the vander with UH&FPO. We hand over the Purchase Order of vander from SCI and GO from DGHS and ask the authority to cooperate the installation and commmissioning of the oxygen manifold system at UHC. As the UHCs are in the Upazila level and the not suitable for outside people to stay other than the hospital area, the vander have taken the formal permission to stay

their TWG at hospital premises during the entire project work.

As a coordinator of the project work I always keep communication with UH&FPO or RMO to ensure smooth working. Also outlets provision for the visited facilities to guide them for making the layout of the pipeline design.

Layout of Manifold Room and Oxygen Pipeline Routes:

During our joint visit for handover we ensure the layout of the manifold room for oxygen manifold system installation. Main room inside dimension is 10'x20'. But along with the ramp and stair about 350 sq-ft areas required with an approach road required to loading and unloading of the oxygen cylinder.



Barhatta UHC Oxygen Manifold Layout



Taranda UHC Oxygen Manifold Layout



Kalmakanda UHC Oxygen Manifold Room Layout



Faridganj UHC Oxygen Manifold Room Layout

Nabinagar Oxygen Manifold Room Layout has given and the construction of foundation of all 4 columns was completed and even 2 column was casting till EGL. But later on stopped after getting objection of MP



Nabinagar Oxygen Manifold Room Layout after Relocation

Pictures: Oxygen Manifold Room Layout of few UHC

Oxygen Manifold Room is a structure required to provide the layout in the selected location at each UHC. The layout of the Oxygen Manifold room has given on the selected location in 20 UHC. But some of the location required being shift due to unavoidable reason. Such as, in Shanthia Upazila Health Complex, the medical waste has found just under the top soil. And it found almost all 4 footing area. During our visit for checking the casting we have shift the location.



Photo: Cancellation of the previous site

Photo: Layout given in New Selected site

Photos: First Selected Site at Shanthia UHC cancelled after found dumping of hospital waste

We also fixed the pipeline layout and ensure the route as per HTM 2022 guideline. Though, we have scheduled them to provide us the initial pipeline layout. But the vendor did not submit within one and half month period has not been hand over to us. Only few sides, we are able to maintain hand sketch layout for maintaining the pipeline layout as per standard HTM 2022.

And all the valves location, Alarm location and outlet positioning was determined by discussion with the duty doctors and nurses. Later on we are able to change some valve location. But still few valves are really in some awkward height. And during the emergency it will really difficult to operate. On these cases we show the TWG of hospital to maintain some tools for standard safety protocol.

Material Check and Inspection of Oxygen Manifold System:

Prior to commencing the civil construction of the manifold room, all construction materials were thoroughly checked by two teams led by two consultants. These teams visited several sites and ensured the quality of the materials. We would like to acknowledge the valuable assistance provided by the vendors, as well as the Upazila Health & Family Planning Officer (UH&FPO) and the MaMoni representative of the respective Upazila Health Complex (UHC) in this process.

Due to logistical constraints, it was not always possible for us to physically visit the delivery locations

of the construction materials. However, the vendors' representatives were cooperative and provided us with photos and videos of the delivered materials. This allowed us to remotely assess the quality and compliance of the materials with the project requirements. The collaborative efforts of the vendors, UH&FPO, and MaMoni representatives have been instrumental in ensuring that only high-quality materials were used in the construction of the manifold room.

We appreciate the proactive approach taken by the vendors and the support extended by the UH&FPO and MaMoni representatives in ensuring the quality of the construction materials, despite the challenges posed by site visits. This has contributed to maintaining the desired quality standards throughout the construction process.

		
<i>Cement for Plaster</i>	<i>500 W D-formed Rod used in Column, floor, Ramp</i>	<i>Cement for Column and Floor Casting</i>
		
<i>First Class Brick for Wall Construction</i>	<i>Brick Chips</i>	<i>First Class Brick for use in floor casting as brick chips</i>
		
<i>1/2" Stone Chips Used for Base of Column</i>	<i>2.8 cfm Sand for Casting</i>	<i>1.2 cfm Sand for Plaster</i>

Pictures: Construction materials used in Manifold Room Construction

The quality of construction works is paramount, and we understand that random inspections play a crucial role in ensuring adherence to quality standards. To maintain the quality of construction works, our consultant, along with the Upazila Health & Family Planning Officer (UH&FPO) representative and our MaMoni personnel, conducted frequent site visits and inspections. These inspections were conducted in a timely manner to monitor the progress of the construction works and ensure compliance with the specified quality standards.

The random inspections were carried out at various stages of the construction process, including foundation, structural, and finishing works, to identify any deviations from the approved plans,

specifications, and quality requirements. Any issues or concerns identified during these inspections were promptly addressed through corrective measures to maintain the quality of the construction works.

The collaborative efforts of the consultant, UH&FPO representative, and our MaMoni personnel, through regular site inspections and monitoring, have been instrumental in ensuring that the construction works meet the desired quality standards and are in compliance with the project requirements.



Photos: Automatic Manifold of pneumatech brand and the display 3.5" found as per there catalog



Photos: Digital Alarm of pneumatech brand



Photos: Atlas Copco Line Ball Valves with lock system

The Project for Oxygen Manifold System Establishment at 20 UHC and 4 Medical gas connectivity at SCANU Project Completion Report

		
	Photos: Medical Grade Copper Pipes and fittings as per required standard BS EN 13348, degreased seamless copper tube and pipes, used for medical gas pipeline system	
	 Photo: Pneumatech Brand Oxygen Flowmeter with Humidifier	
	Photo: Supplied Oxygen Cylinder inspected and identified the manufacturing date, Capacity, Working Pressure	

Photos: Different supplied material at site

Civil Construction of Oxygen Manifold Room of 20 UHC

Despite being unable to physically visit the site during the footing casting process, we have implemented quality assurance measures through remote monitoring using WhatsApp and video calls. This allows us to closely monitor the progress and quality of the footing casting remotely. We work closely with the contractor and other stakeholders to ensure that the casting process adheres to the required quality standards and specifications. Through continuous communication and visual inspections via WhatsApp and video calls, we are able to address any concerns or issues in real-time, provide necessary guidance, and ensure that the footing casting meets the desired quality standards. This remote monitoring approach enables us to maintain quality control and ensure that the project progresses smoothly, even in challenging situations where physical presence is not possible.

The Technical Working Group (TWG) of SIL has been highly cooperative in maintaining the construction schedule across 20 sites. We have made concerted efforts to ensure quality works by maintaining regular communication with the Upazila Health & Family Planning Officer (UH&FPO), Regional Maintenance Officer (RMO), and local MaMoni representative. Through timely updates and coordination with these stakeholders, we have been able to monitor the progress of the construction works, address any concerns or issues, and ensure that quality standards are met. The collaborative efforts of the TWG, along with effective communication and coordination with relevant stakeholders, have played a pivotal role in ensuring the smooth progress of the construction works and maintaining the desired quality standards throughout the project.



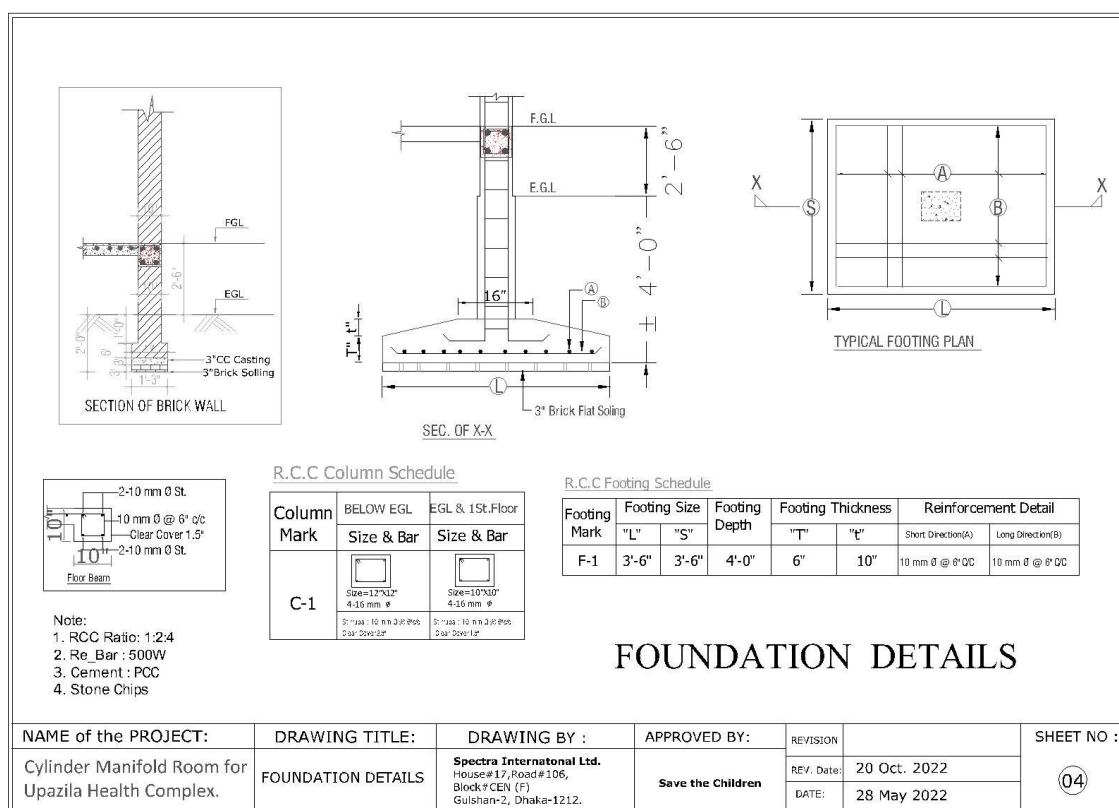
Pictures: Column Footing Casting Process

The RCC casting ratio and other details for the foundation, column, floor, ramp, wall plaster, and mortar has specified before the civil works started. Adhering to the recommended ratios and using appropriate materials, such as stone chips for foundation and brick chips for structure above EGL, is important for ensuring the quality and integrity of the construction.

It is using a ratio of 1:2:4 for RCC casting (i.e., 1 part cement, 2 parts sand, and 4 parts coarse aggregate) for the foundation and column below EGL is a common practice in construction. Similarly, using the same ratio of 1:2:4 for cement concrete (CC) for the floor and ramp is also widely followed. The recommendation to use stone chips as coarse aggregate in the foundation is suitable as they provide good stability and strength to the foundation. Brick chips, on the other hand, are commonly used in structures above the EGL as they can offer better aesthetics and are lighter in weight compared to stone chips.

The wall plaster thickness of 12mm and the mortar ratio of 1:6 (cement to sand) are also standard practices in construction. Proper plastering of walls can help protect them from moisture, improve aesthetics, and enhance durability. Adhering to the recommended mortar ratio is essential for ensuring the right consistency and strength of the plaster.

Maintaining strict quality control throughout the construction process, including the use of recommended ratios, materials, and construction techniques, is vital for ensuring the structural integrity and durability of the building. Regular monitoring and inspection can help identify and address any issues or deviations from the recommended standards, leading to a successful construction project.



Initially the manifold room has no column structure, later on considering the earthquake hazzard region at Sylhet, Netrokona, Mymensingh and Brahmanbaria we recommended the column structure to sustain the stucture against the sysmic load.

We have visited al 20 UHC during our construction works at different stages to maintain the quality work along with to mitigate different challenges at different sites.

 Photo: Earth Excavation Work at Kanaighat UHC	 Photo: Brick laying for subsoil structure Beanibazar UHC
 Photo: Foundation Casting Narria UHC	 Photo: Column Casting Shibchar UHC
 Photo: Column Casting and subsoil wall at Begumganj UHC	 Photo: Beam and Floor rod fabrication works at Bhedergonj UHC
 Photo: Floor Casting at Shanthia UHC	 Photo: Beam, Floor and Ramp after Casting at Nazirpur UHC

Pictures: Different Civil Construction Works Satages of Oxygen Manifold Room at Different UHC



Photo: Brick works for Oxygen Manifold Room



Photo: Brick works for Oxygen Manifold Room



Photo: MS Truss works for Oxygen Manifold Room



Photo: Complete Oxygen Manifold Room at Naria UHC



Photo: Oxygen Manifold Room Hatiyah UHC before colour work



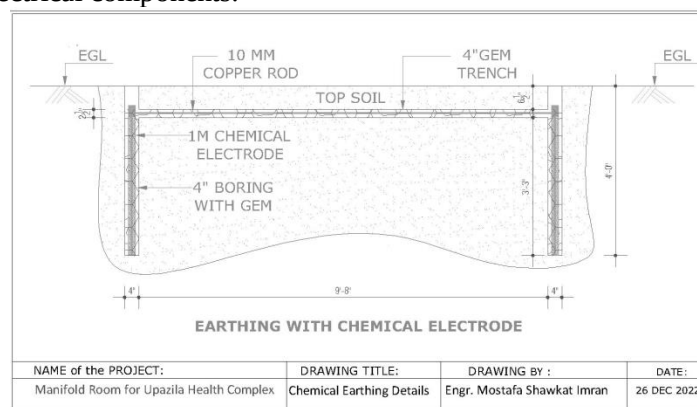
*Photo:
Oxygen
Manifold
Room
Hatiyah UHC
after colour
work*

Photos: Different stages of construction works at different UHC

Electrical Works quality check along with Earthening works

In this project, the tender specification did not initially include electrical works. However, in order to ensure the proper operation of the oxygen manifold room, we recommend installing a lighting system that provides adequate illumination for the room. Additionally, we suggest installing two ceiling fans

and one exhaust fan to maintain a comfortable environment inside the room. We also emphasize the importance of implementing a robust earthing system for all electrical appliances, including the Automatic Manifold and lightning arrestor earthing for the manifold room, to ensure safety and proper functioning of the electrical components.



After the procurement process, it has come to our attention that electrical works were not included in the project. However, it is crucial to provide power supply for the operation of the automatic manifold, as the connectivity of oxygen to the hospital is a critical requirement. Furthermore, the manifold room is a newly constructed structure, and therefore, power supply becomes essential for its functioning.

Additionally, the structure of the manifold room is made of steel and has a tin shed on its roof. It is located outside and separate from the main hospital building. Considering these factors, it is imperative to incorporate electrical works into the project scope to ensure the proper functioning and operational readiness of the oxygen manifold and related facilities, as well as comply with safety and regulatory requirements. This may involve coordination with vendors, contractors, and relevant stakeholders to efficiently address the electrical requirements of the manifold room.



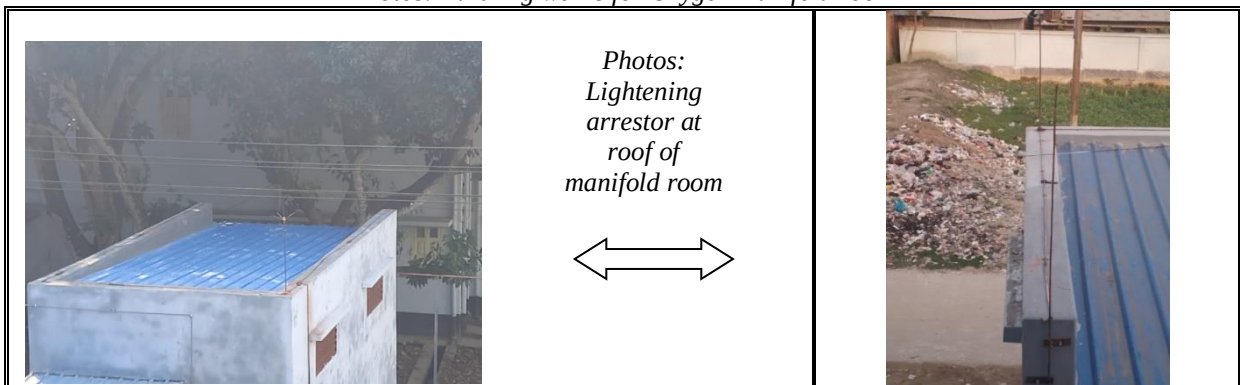
Photos: Eelectrical works for Oxygen manifold room

To mitigate the risk of lightning strikes, we recommend implementing a separate earthing system for the structure. We propose using Ground Enhancement Material (GEM) to improve the grounding system. Typically, a 10 mm diameter copper rod is conventionally used along with other materials to

achieve the desired ground resistance level. However, using GEM has shown to yield superior results. The copper electrode rod is embedded in the GEM material, and a galvanized, protected 50 mm pipe is used as the electrode. This configuration has been found to achieve a ground resistance level of below 3 ohms, ensuring improved performance and enhanced protection against lightning strikes.



Photos: Earthing works for Oxygen manifold room



*Photos:
Lightening
arrestor at
roof of
manifold room*

Pipeline brazing with Nitrogen purging

The installation of copper pipelines for medical gas pipelines involves special arrangements for brazing. Oxygen and acetylene are used for brazing the joints with Silver-brace soldering, in the presence of nitrogen gas inside the copper pipelines. The nitrogen gas creates an inert gas atmosphere inside the copper pipes, preventing the formation of unburned carbon on the copper pipe surface.

In our project, SIL considerations led us to initially avoid brazing works using nitrogen gas purging. However, upon inspection at Southsurma UHC, we discovered that no nitrogen gas purging was being used during copper pipe brazing works. As a result, we immediately halted all brazing works at all sites. At South Surma, we demonstrated the carbon formation inside the copper pipe surface by cutting a joint where nitrogen gas was not used during brazing. We shared a video in our WhatsApp working group and instructed all site managers to prioritize nitrogen gas purging at all costs. Upon realizing this issue, the Technical Working Group (TWG) of SIL ensured that nitrogen gas purging was maintained in all other sites during brazing works.



Photos: Save the Children Team visited the Beanibazar site during the pipeline works brazing



Photos: Nitrogen gas purging during brazing works at different sites

Pipeline System layout and installation:

In our country the medical gas pipeline installation work has followed eighter MTM 2022 or NFPA 99. And we recommended the vendor to follow the all required guidline from design to installation.



Photo: Marking and drilling

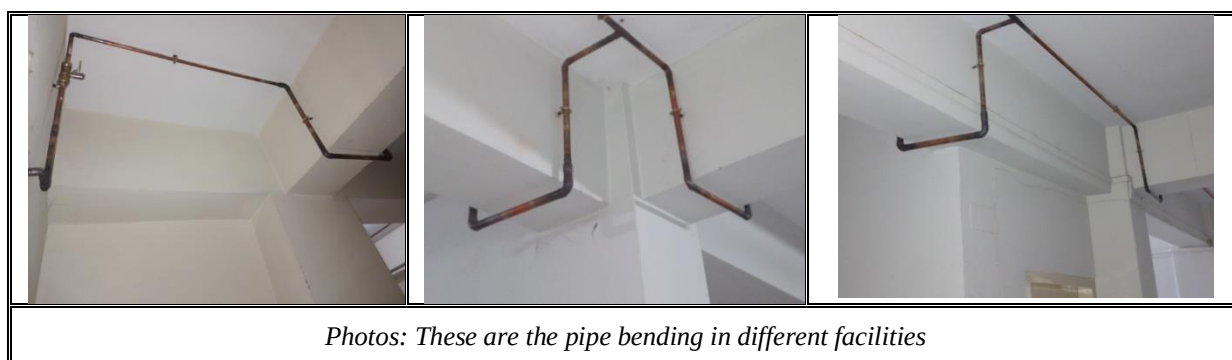
Photo: Marking and drilling



Photos: Oxygen Pipeline Installation as per recommended standard

Though some places we have found the pipeline installed in the ceiling and bended up and down too much where the beam has been suspended from the ceiling. Some places they have changes as per our recommendation but some place they did not.

The valves should install as per the HTM 2022, where it was recommended to installed in the permissible height where a nurse or medical staff could control it during emergency shut-off required. The maximum line ball valves were installed in suitable height, where as some installed in very awkward location and height.





The vendor's explanation for such installation was, as oxygen pipeline system in the upazila health complexes are not very familiar and may cause sabotages or malhandling during the operation. And it would be reduce the accident probabilities. So, we accept such installtion to avoid the accident

Safety and security of the total system is highly recommended. The operation and maintenance training to control of medical oxygen and frequent promotional activities to generate the awareness program for medical oxygen uses with medical gas outlet among the patients and their attendance in the local community could reduce the risk of accident.

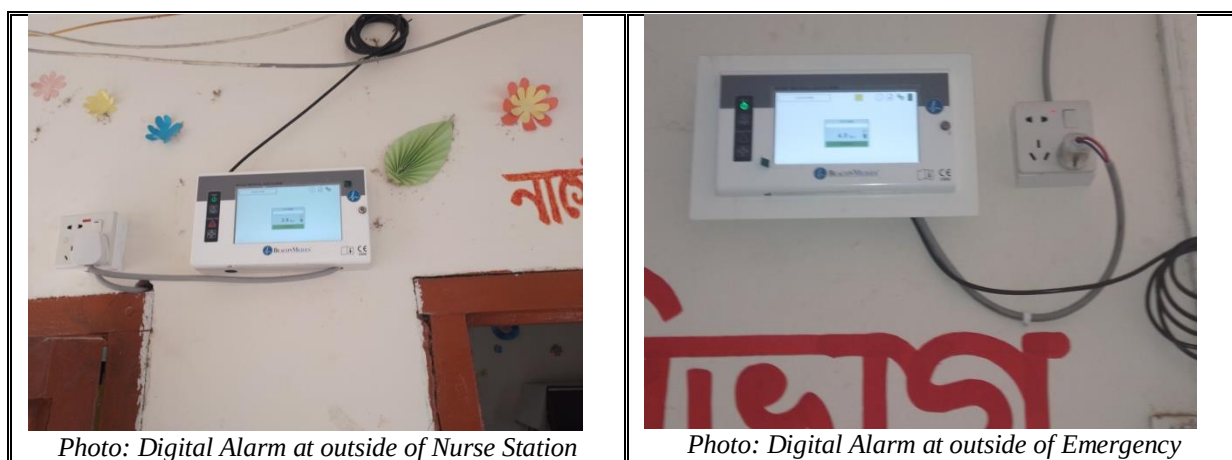




Photo: Digital Alarm at outside of OT



Photo: Digital Alarm Calibration by SIL Engineer

Installing digital alarms in the operation theatre and emergency areas of each UHC is a good practice to ensure proper monitoring and control of the oxygen manifold system. These alarms can help in detecting any deviations in line pressure and alerting the relevant personnel for timely action.

Calibration of the digital alarms is important to maintain accurate readings and ensure that the alarms are triggered at the appropriate pressure levels. Based on the information provided, it seems that the digital alarm is currently set to trigger a lower level audible alarm at 3.5 Bar (which is below the actual line pressure of 3.9 Bar) and a higher level audible alarm at 4.5 Bar.

To maintain balance pressure with the manifold, it is recommended to calibrate the digital alarms to the desired pressure levels based on the system requirements and specifications. This may involve adjusting the settings of the digital alarms to match the actual line pressure readings, so that the alarms are triggered at the intended pressure levels. Proper calibration of the alarms can help ensure that any deviations in line pressure are promptly detected and addressed, ensuring the safe and efficient operation of the oxygen manifold system at the UHC. It's important to follow manufacturer's guidelines and local regulations while calibrating the digital alarms.

Testing and Commissioning of Oxygen Manifold System:

Performing leakage testing on the Oxygen Manifold System is a crucial step to ensure the integrity and safety of the system. The procedure you described, using Dry Nitrogen Gas at 180% of the working pressure (i.e., 7 bars for a system with a working pressure of 4 bars) and monitoring for 24 hours, is a common practice in the industry.

By pressurizing the system with Dry Nitrogen Gas at a higher pressure than the working pressure, any potential leaks can be easily identified. Monitoring the pressure over a 24-hour period allows for sufficient time to detect any changes in pressure that may indicate leakage.

The use of soap foaming at every joint is a reliable method to identify leaks. If any leakage is detected, re-soldering or other appropriate measures should be undertaken to fix the issue and ensure a leak-proof oxygen line.

It's important to follow proper safety protocols and guidelines during the leakage testing process, including using appropriate PPE (Personal Protective Equipment), working in well-ventilated areas, and adhering to local regulations and standards. Regular inspection and monitoring of the system during and after the leakage testing process can help identify and address any issues, ensuring that the Oxygen Manifold System is safe and reliable for use in medical gas supply.



Commissioning of Oxygen Manifold System at 20 UHC:

As the coordinator of the project, I am pleased to report on the successful commissioning of the oxygen manifold system at 20 UHCs (Upazila Health Complexes) as part of our project implementation. The commissioning process involved several key steps to ensure the proper installation, testing, and calibration of the system to meet the required standards for medical gas pipeline work and safety protocols.

The project team worked closely with the vendor and subcontractors to ensure that the construction of the manifold room and copper pipeline brazing works were carried out simultaneously, as per the revised work plan based on the logical explanation provided by the vendor. The quality of the construction, including the RCC casting ratio for foundation and columns, and the cement concrete ratio for floors and ramps, was strictly maintained, with appropriate use of stone chips and brick chips as coarse aggregates based on the designated areas of the structure.

The commissioning process also involved thorough testing of the oxygen manifold system for leakage using dry nitrogen gas at 180% of the working pressure, and monitoring the pressure for 24 hours to ensure that it did not decrease. Any leakages detected were promptly addressed through re-soldering of the oxygen lines. Digital alarms were also installed in the operation theatre and emergency areas of each UHC, calibrated to appropriate pressure levels for triggering audible alarms in case of deviations

in line pressure.

Regular communication was maintained with UH&FPO (Upazila Health and Family Planning Officer) and RMO (Residential Medical Officer) to ensure smooth working and provide guidance for layout design of the pipeline. The successful commissioning of the oxygen manifold system at 20 UHCs is a significant milestone in our project, ensuring a safe and reliable supply of oxygen to support the healthcare facilities in remote Upazilas. We thank all stakeholders, including the vendor, subcontractors, UH&FPO, RMO, and the project team for their dedication and efforts in achieving this milestone. Moving forward, we will continue to monitor and maintain the oxygen manifold system to ensure its safe and efficient operation in accordance with the specified standards and protocols of Save the Children and DGHS (Directorate General of Health Services).

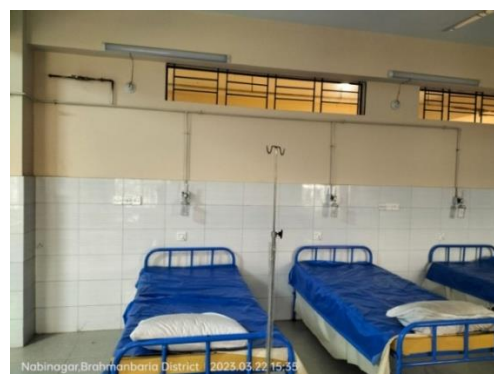


Photo: Installation of Manual Manifold (2X6)



Photo: Commissioning of Manual Manifold (2X6)



Photo: Additional Air provided at Nabinagar UHC



Photo: Inaguration at Mathbaria UHC

Commissioning of Medical Gas Connectivity of SCANUs:

Commissioning of Medical Gas Connectivity of SCANU at Sylhet Osmani Medical College Hospital

The SCANU (Special Care Newborn Unit) at Osmani Medical College Hospital funded by USAID and built by Save the Children, faced challenges during the COVID-19 pandemic due to the lack of Medical Air and Vacuum lines. With only 36 sets of US standard Ohmeda style Medical Oxygen

outlets, the SCANU was unable to fully function, as Medical Air outlets were required to drive Bubble CPAP machines and high flow nasal cannulas, and a Medical Vacuum system was also urgently needed.

Initially, the recommended number of outlets was 60 sets of Medical Vacuum, and 20 sets each of Medical Oxygen and Medical Air. However, due to space constraints in the existing building, it was not feasible to install all 100 sets of outlets as planned. To address this challenge, a rearrangement of outlets was proposed, introducing Bed Head Units as per standard protocols, to optimize space utilization and provide maximum facilities in each unit at SCANU.

The installation of the Medical Air and Vacuum lines also required an underground connectivity from a new building, where the Vacuum and Air plant was installed. The project team worked diligently to ensure compliance with safety standards and protocols during the installation process.

Through collaborative efforts, the commissioning of Medical Air and Vacuum lines was successfully completed, providing the SCANU with the necessary infrastructure for optimal patient care. We express our gratitude to Save the Children, USAID, the project team, and hospital authorities for their support in overcoming the challenges and ensuring the installation of Medical Air and Vacuum lines at SCANU of Osmani Medical College Hospital.





Photos: Commissioning of 100 outlets at Sylhet Osmani Medical College Hospital Medical Gas connectivity works

Commissioning of Medical Gas Connectivity of SCANU and maternity wards at Habiganj 250 Bed Sadar Hospital

The commissioning of the Medical Gas Pipeline System at Habiganj Sadar Hospital was completed to cater to the requirements of 14 SCANU seats, as well as other wards including Emergency, Maternity, Labour room, and Observations.

A total of 14 sets of Medical Oxygen, Medical Air, and Medical Vacuum outlets were installed to serve the SCANU seats, ensuring that adequate medical gas supply is available for critical care needs. Additionally, Medical Oxygen and Medical Vacuum outlets were also distributed among the beds in other wards to support patient care.

The Medical gases were sourced from the Oxygen mainline and Air and Vacuum plant at Prime Sheikh Hasina Medical College hospital building, where the Medical Gas Pipeline System was installed earlier. To ensure seamless connectivity, an underground trench was made as per the guidelines of HTM 2022 to cross the road, ensuring compliance with safety and quality standards.

The commissioning of the Medical Gas Pipeline System at Habiganj Sadar Hospital was executed with meticulous planning, adhering to best practices and guidelines. The project team worked diligently to ensure proper installation, testing, and calibration of the system to ensure safe and reliable medical gas supply to meet the healthcare needs of patients in various wards, including the critical SCANU unit. We extend our appreciation to all stakeholders involved in the successful commissioning of the Medical Gas Pipeline System at Habiganj Sadar Hospital.

We express our gratitude to Save the Children, USAID, the project team, and hospital authorities for their support in overcoming the challenges and ensuring the installation of Medical Air and Vacuum lines at SCANU of Habiganj 250 Beds Sadar Hospital.



Photos: Commissioning of 100 outlets at Habiganj Hospital Medical Gas connectivity works

Commissioning of Medical Air connectivity of SCANU at Sunamganj 250 Bed Sadar Hospital

The commissioning of medical gas connectivity works at Sunamganj 250 Bed Sadar Hospital was completed to cater to the requirements of SCANU, which included Medical Oxygen, Medical Air, and

Medical Vacuum outlets. The installation of Medical Oxygen and Medical Vacuum outlets was carried out by PWD during the COVID-19 pandemic project. However, only 14 Medical Air outlets were installed initially in the SCANU.

In addition to the medical gas outlets, Save the Children International also provided other instruments for the SCANU, and installed the air conditioning systems. As part of our technical consultancy, we identified a potential electrical hazard due to the co-existence of the electrical consoles at the same height as the oxygen gas outlets, posing a risk of fire hazard. To mitigate this risk, we recommended and executed the re-positioning of the electrical consoles to a lower level, aligning them with the outlet level to eliminate the hazard.

The commissioning of medical gas connectivity works at Sunamganj 250 Bed Sadar Hospital was carried out with utmost consideration for safety and compliance with relevant regulations and standards. The successful completion of the project was made possible through collaborative efforts between different stakeholders, including PWD, Save the Children International, and our team providing technical consultancy. We are committed to ensuring the safety and reliability of medical gas supply in healthcare facilities, and commend all parties involved in the successful commissioning of the project.

We express our gratitude to MaMoni of Save the Children, USAID, the project team, and hospital authorities for their support in overcoming the challenges and ensuring the installation of Medical Air at SCANU of Sunamganj 250 Beds Sadar Hospital.



Photos: Commissioning of 14 outlets at Sunamganj Hospital Medical Gas connectivity works

Commissioning of Medical Gas Connectivity of SCANU at Manikganj 250 Bed Sadar Hospital

The commissioning work for medical gas connectivity at SCANU of Manikganj Sadar Hospital was completed despite facing several challenges. Unlike other SCANUs where underground cross road connections were possible, in this case, an overhead structure had to be provided to transfer the medical gas pipeline from the new building where the gas pipeline system was installed. This presented a unique challenge in terms of structural design and installation.

Additionally, the existing gas pipelines were found to have multiple bends and varying heights, leading to pressure drops during operational mode at SCANU. To address this issue, the pipelines had to be re-installed to align at a single height, just below the beam height, ensuring smooth and efficient gas flow to the SCANU.

Furthermore, the configuration of outlets in the SCANU was different from other sites, with only 3 Oxygen outlets and 1 Medical Air and Vacuum outlet for each bed. This required careful planning and coordination to ensure that all 100 sets of outlets were accommodated in the SCANU while adhering to the standards and guidelines for medical gas supply.

Despite these challenges, the commissioning of medical gas connectivity works at Manikganj Sadar Hospital SCANU was successfully completed, thanks to the expertise and dedication of the team involved. The project underscores our commitment to overcoming obstacles and delivering safe and reliable medical gas supply solutions in healthcare facilities.

We express our gratitude to MaMoni of Save the Children, USAID, the project team, and hospital authorities for their support in overcoming the challenges and ensuring the installation of Medical Air and Vacuum lines at SCANU of Manikganj 250 Beds Sadar Hospital.





Photos: Commissioning of 100 outlets at Manikganj Hospital Medical Gas connectivity works

Operation and Maintenance Training at 20 UHC:

Training Session on Operation and Maintenance of Oxygen Manifold and Equipment

Objective of the training : The training session aimed to educate junior mechanics, ward boys, senior staff nurses, doctors, and UH&FPO/RMO personnel on the proper operation and maintenance of oxygen manifolds, manual manifolds, digital alarms, line ball valves, and outlets, as well as the use of flow meters and humidifiers.

Equipment Covered:

1. **Oxygen Automatic Manifold:** An oxygen automatic manifold is a system that automatically distributes oxygen from a central source to multiple outlets. It is used to supply oxygen to various areas in a hospital or healthcare facility. The proper operation and maintenance of the manifold were explained, including how to check for leaks, ensure proper pressure, and monitor alarms for any issues.
2. **Manual Manifold:** A manual manifold is a system that requires manual adjustments to control the flow of oxygen to different outlets. The training covered how to properly set the flow rate, monitor pressure, and check for any blockages or leaks.
3. **Digital Alarms:** Digital alarms are used to monitor and alert personnel to any abnormalities in the oxygen supply system. The training session covered how to interpret and respond to different alarms, such as low oxygen pressure, high pressure, or flow rate deviations.
4. **Line Ball Valves:** Line ball valves are used to control the flow of oxygen in the pipelines. The training included instructions on how to properly open and close the valves, as well as how to identify and address any issues with the valves.
5. **Outlets:** Oxygen outlets are used to connect various devices, such as oxygen masks or ventilators, to the oxygen supply system. The training covered how to properly connect and disconnect the outlets, as well as how to clean and maintain them.
6. **Flow Meters along with Humidifiers:** Flow meters are used to measure the flow rate of oxygen in the pipelines. The training session included instructions on how to properly

read and interpret flow meter readings, as well as how to calibrate and maintain the flow meters.

Humidifiers are used to add moisture to the oxygen supply to prevent drying of the respiratory system. The training covered how to properly fill, clean, and maintain the humidifiers, as well as how to monitor and adjust humidity levels.

Key Points:

1. Proper operation and maintenance of oxygen manifolds, manual manifolds, digital alarms, line ball valves, outlets, flow meters, and humidifiers are essential to ensure a continuous and safe oxygen supply in a healthcare facility.
2. Regular checks for leaks, pressure monitoring, and alarm interpretation are important to identify and address any issues with the oxygen supply system promptly.
3. Proper training should be provided to all personnel involved in the operation and maintenance of oxygen equipment, including junior mechanics, ward boys, senior staff nurses, doctors, and UH&FPO/RMO personnel.
4. Compliance with manufacturer's instructions, hospital protocols, and safety guidelines is crucial to ensure the safe and effective operation of oxygen equipment.
5. Any deviations from normal operation or alarms should be reported immediately to the appropriate personnel for timely resolution and to avoid disruptions in oxygen supply.

The training session provided by SIL engineers aimed to educate junior mechanics, ward boys, senior staff nurses, doctors, and UH&FPO/RMO personnel on the proper operation and maintenance of oxygen manifolds, manual manifolds, digital alarms, line ball valves, outlets, flow meters, and humidifiers. The training emphasized the importance of regular checks, alarm interpretation, and compliance with manufacturer's instructions and hospital protocols to ensure a continuous and safe oxygen supply in a healthcare facility. By following proper procedures and guidelines, the personnel can effectively operate and maintain the oxygen equipment, contributing to patient safety and care



Photo: Taining session at Mathbaria UHC



Photo: Taining session at Kachua UHC



Photo: Taining session at Kachau UHC



Photo: Taining session at Kachau UHC



Photo: Taining session at Mathbaria UHC



Photo: Training on operation and maintenance of Automatic Manifold at Kachua UHC by Engr. Md. Kamal Hossain

4. CONCLUTIONS

Summery of Project results:

The summery of the project results and outcomes of the Medical Oxygen Manifold System Project implemented by Save the Children International in 20 Upazila Health Complexes in Bangladesh, funded by USAID, can include:

1. Establishment of Oxygen Manifold Systems and Medical Gas Connectivity: The project successfully procured, installed, and commissioned oxygen manifold systems in 20 Upazila Health Complexes, ensuring a reliable and continuous supply of medical oxygen to patients in need. Specially the children and the mothers of Bangladesh at rural areas. NICU at UHCs and SCANU of 4 hospitals get previlages and ensure smooth operation of critical care services of newborn babies of those areas.
2. Improved Access to Medical Oxygen at Upazila Level and SCANU at District Hospitals: The project contributed to improved access to medical oxygen for patients, specially newborns and maternity with respiratory conditions, including COVID-19, by ensuring a steady supply of oxygen in the health facilities. This resulted in timely and appropriate oxygen therapy for patients, potentially reducing mortality rates and improving patient outcomes.
3. Enhanced Patient Care and Outcomes: Health personnel were trained on the use and maintenance of the oxygen manifold systems, enabling them to provide high-quality oxygen therapy to patients, alleviate respiratory distress, and improve oxygen saturation levels. This contributed to enhanced patient care and outcomes, with improved clinical outcomes, reduced complications and better recovery rates for patients, especially new borns and mothers.
4. Strengthened Health System Capacities: The project provided capacity building and training for health personnel on infection prevention and control measures, contributing to strengthened health system capacities. Health workers were equipped with the skills and knowledge to effectively manage patients requiring oxygen therapy and respiratory care, enhancing the overall capacity of the health system.
5. Benefits to Communities and General Public: The project's outcomes had a positive impact on the communities and the general public served by the 20 Upazila Health Complexes and the newborns of 4 SCANUs of 4 different districts. Improved access to medical oxygen and enhanced patient care contributed to better health outcomes and well-being of the communities, reducing the burden on patients and their families, and promoting better overall health in the project areas.

Findings and Recommendation of the projects:

Findings:

1. Oxygen Manifold System Establishment: The establishment of an oxygen manifold system in 20 UHC and as part of the Medical Gas Connectivity for SCANU project was found to be crucial in ensuring a reliable and efficient oxygen supply to support newborn care. The manifold system provided a centralized and controlled distribution of oxygen to multiple

outlets, facilitating the effective management of oxygen supply in these healthcare facilities.

2. **Improved Oxygen Supply:** The implementation of the oxygen manifold system was found to significantly improve the oxygen supply in the UHCs, ensuring a consistent and adequate supply of oxygen to meet the needs of newborns requiring respiratory support. The manifold system helped to regulate the flow of oxygen, monitor pressure, and control the distribution of oxygen, minimizing interruptions and ensuring a continuous supply of oxygen for newborn care.
3. **Enhanced Patient Care:** The establishment of the oxygen manifold system positively impacted patient care in the UHCs by ensuring that newborns in need of oxygen therapy received timely and appropriate care. The reliable oxygen supply facilitated the effective use of oxygen therapy in newborns with respiratory distress, improving their chances of survival and reducing the risk of complications.
4. **Centralized Monitoring and Control:** The implementation of the manifold system allowed for centralized monitoring and control of the oxygen supply, enabling healthcare personnel to quickly detect and address any issues or deviations in oxygen pressure, flow rate, or alarms. This centralized control facilitated prompt troubleshooting and maintenance, minimizing downtime and ensuring uninterrupted oxygen supply.
5. **Ensuring Quality Supplier and Installer:** Though Spectra International Limited is one of the leading stakeholders in this sector. They need to realize to design the medical gas pipeline system is the key factor to ensure the quality service in these sectors. They have the experts but lack of standard practice may cause damage their name and fame in this sector in the near future. Beside this to find out the quality supplier as well as contractor well experienced consultant has to be appointed from the beginning of the project like this.

Recommendations:

1. **Regular Maintenance and Monitoring:** It is recommended to establish a regular maintenance and monitoring schedule for the oxygen manifold system in the UHCs. This should include routine checks for leaks, pressure monitoring, alarm verification, and flow meter calibration. Proper documentation of maintenance activities should be maintained for reference and audit purposes.
2. **Staff Training:** Adequate training should be provided to healthcare personnel, including doctors, nurses, and technicians, who are responsible for operating and maintaining the oxygen manifold system. This should include training on the proper operation of the manifold system, interpretation of alarms, troubleshooting, and emergency response procedures.
3. **Compliance with Standards and Guidelines:** The oxygen manifold system should be installed, operated, and maintained in compliance with relevant standards like BNBC, HTM 2022, NFPA 99, guidelines, and regulations, such as those issued by the local health department, medical gas suppliers, and other governing bodies. Regular audits and inspections should be conducted to ensure compliance.
4. **Emergency Preparedness:** Emergency preparedness plans should be in place to address any unforeseen interruptions in the oxygen supply. Like the leakage of oxygen at terminal unit may cause fire hazard, because we have found the rural people has lack of knowledge and

safety protocol to maintain standard practice the helath hazard area. They frequently smoke inside the wards, where the patients are most infection in the respiratory track infection. This should include backup oxygen supply options, contingency plans for equipment failure, and emergency response protocols to ensure the safety and well-being of patients.

5. **Documentation and Reporting:** Proper documentation of the oxygen manifold system's operation, maintenance, and monitoring activities should be maintained, including records of pressure readings, alarm logs, and maintenance schedules. The flow meters are found some leakage at the junction of humidifiers. So, properly reporting in this issue could save a lot of operating cost of oxygen. Reporting of any issues or deviations should be timely and accurate to facilitate prompt resolution.
6. **Regular Review and Improvement:** The oxygen manifold system should be regularly reviewed for its effectiveness and efficiency, with opportunities for continuous improvement identified and implemented. Feedback from healthcare personnel and stakeholders should be sought and used to enhance the system's performance and address any shortcomings. Such as there is no medical gas monitoring system (Alarm or Presure Guage) at SCANU system. So, in next up coming improvement the 3 gas Alarms or digital monitor along with central BMS system could be upgraded at all 4 SCANUs.

In conclusion, the establishment of an oxygen manifold system in 20 UHCs as part of the Medical Gas Connectivity for SCANU project has been found to be crucial in improving oxygen supply and enhancing patient care. Proper maintenance, staff training, compliance with standards, emergency preparedness, documentation, and regular review are recommended to ensure the efficient and reliable operation of the manifold system and to provide optimal care to newborns in need of oxygen therapy.

Application and Acknowledgement:

Application

In this report I have tried to investigate the overall activities of the contractors as they are the largest company who has implemented the medical gas pipeline systems at major governemnt owned hospitals and healthcare facilities in Bangladesh. They are also one of the biggest Oxygen suppliers in Bangladesh. This report focus on some malpractices that could lead loses of different resources in this sector. We tried to successfully complete the project by mitigate all negative approaches and we will share this report with all relevant stakeholders, such as project partners, donors, government authorities, and other interested parties. This report would be utilized for future planning, decision-making, and implementation of similar projects or interventions. And hope found some key points to review their current decision.

Acknowledgement

"I would like to begin by expressing my gratitude towards Allah for giving me the opportunity to work on this meaningful project, where I could utilize my expertise to serve humanity during these challenging times.

I am deeply grateful to Save the Children International and USAID, two remarkable organizations that have played a vital role in our healthcare sector. The realization of the need for oxygen at the grassroots level during the COVID-19 pandemic was the driving force behind this project, which was indeed a costly undertaking. Without the generous financial support of USAID, it would not have been possible to effectively complete the project. Their unwavering commitment and extended budgetary support have ensured the smooth implementation of the project. I am thankful to everyone in these organizations for their amiable demeanor and big-heartedness.

I would like to express my heartfelt appreciation to my previous Project Director, Dr. Md. Khairul Alam, who has provided tremendous support whenever I have approached him. His leadership skills and team-building efforts have infused me with extra motivation to achieve our project goals. Without his kind cooperation, it would have been challenging to accomplish what we have. He accommodated all required facilities to smoothly drive our activities effectively.

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